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Selecting Optimum Planting Dates and Plant Populations for Dryland Corn in Kansas

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Research Question

Corn production practices have to be evaluated periodically to determine whether management changes are needed to fully utilize improvements in water and nutrient use efficiency as well as shade, drought, cold, and pest tolerance. Increased tolerance to shade and drought stress should enable producers to increase plant populations, resulting in higher yields in the more productive years without the risk of yield loss during a drought year. Greater cold tolerance and seedling vigor should allow earlier planting dates, thus enabling producers to take advantage of fuller season hybrids. The objectives of this study were to evaluate the effects that plant population, planting date, and hybrid maturity have on corn yield.

Literature Summary

Studies conducted in 1966 and 1967 in Kansas indicated that dryland corn yields increased as plant population increased until reaching a maximum at approximately 20 000 plants/acre and then declined as plant population continued to increase. However, recent studies have reported that the corn yield response to plant population was quadratic with yields reaching a plateau rather than declining at higher plant populations. Corn producers in northeast Kansas may plant earlier to increase the growing season length or to move harvest earlier in the year to distribute labor. Approximately 260 heat units can be accumulated during April in northeast Kansas, yet little data exist to quantify the effects that these additional heat units have on corn yields. As corn acres increase in Kansas, management decisions must be modified to fit the hybrids currently being used to the environment.

Study Description

A 2-yr corn study was conducted at three sites in Kansas with the following experimental design:

Main Plots: Planting dates of approximately 1 April, 1 May, and 1 June

Subplots: Target populations of 14 000, 20 000, and 26 000 plants/acre and two hybrids of 102-d and 113-d relative maturities.

Applied Questions

Does delaying planting from early April until early May or June reduce grain yields in northeast Kansas?

At five of the six location-years, planting in early April and early May resulted in similar grain yields. Higher temperatures and lower rainfall amounts in June reduced grain yields with an April planting compared with a May planting date at one location-year. Delaying planting until early June reduced grain yields on average 36 bu/acre across all six location-years.

What is the impact of plant population on corn yields in Northeast Kansas?

Increasing corn plant population increased grain yields under a majority of the environments and reduced grain yields under one set of very low yielding conditions. The average grain yield at 26 000 plants/acre was 118 bu/acre, whereas

Full scientific article from which this summary was written begins on page 85 of this issue.

grain yields were 113 bu/acre at 20 000 plants/acre and 98 bu/acre at 14 000 plants/acre across all environments.

As planting is delayed in northeast Kansas, should a short-season corn hybrid be planted?

Grain yield response to hybrid maturity varied consistently with planting dates across all three locations. As expected, the full-season hybrid produced 23 bu/acre more grain than the short-season hybrid when planted in April. The difference between the two hybrids was reduced to approximately 4 bu/acre when they were planted in May. When planted in June, the short-season hybrid produced yields that averaged 5 bu/acre higher than those from the full-season hybrid.

Our results indicate that if early planting dates are used, producers should take advantage of the longest season hybrid adapted for the area. If planting is delayed to early June, using a short-season corn hybrid appears to have an advantage. In half of the six location-years, the short season hybrid produced 4 bu/acre less grain than full season hybrid when planted in June. However, in the location-years when the short-season hybrid produced higher yields than the full-season hybrid with a June planting, the average yield difference was 22 bu/acre.

Selecting Optimum Planting Dates and Plant Populations for Dryland Corn in Kansas

S. A. Staggenborg,* D. L. Fjell, D. L. Devlin, W. B. Gordon, L. D. Maddux, and B. H. Marsh

Corn (*Zea mays* L.) production practices have to be evaluated periodically to ensure that producers are fully using improvements in hybrids. A 2-yr study was conducted to assess corn yield response to plant population, planting dates, and hybrid maturity. The study was conducted at three sites during the 1994 and 1996 growing seasons. A 102 and a 113 d relative maturity corn hybrid were established at plant populations of 14 000, 20 000, and 26 000 plants/acre. Planting dates in early April, May, and June were also used. At two locations, delaying planting from April to May decreased yields slightly. At the third location, yields increased 27 bu/acre as planting was delayed until early May. Delaying planting at this location resulted in ear development occurring after a period of severe drought, which reduced yields with the April planting. At all three locations, delaying planting until early June reduced yields as a result of ear development under higher temperatures and grain fill occurring under cooler temperatures. Increasing population from 14 000 to 20 000 plants/acre resulted in a yield increase of 14 bu/acre across all six environments. Grain yields increased an additional 4 bu/acre when plant population was increased to 26 000 plants/acre. When planted in April or May, the full season hybrid produced higher yields than the earlier maturing hybrid. Corn producers in north central and northeast Kansas can increase corn yields by using planting dates in April and a population of approximately 26 000 plants/acre.

CORN PRODUCTION practices have to be periodically evaluated to determine whether management changes are needed to fully utilize new information, technology, and hybrids. Improvements in water and nutrient use efficiency as well as shade, drought, and pest tolerance may allow for higher seeding rates and yields. Greater cold tolerance and seedling vigor will allow earlier planting dates, thus enabling producers to take advantage of fuller season hybrids.

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Studies conducted in 1966 and 1967 in Kansas indicated that corn yields increased as plant population increased until grain yields reached a maximum and then declined as plant population continued to increase (Vanderlip, 1968). Recent studies have shown that corn yield response to plant population was quadratic, with yields reaching a plateau rather than declining at higher plant populations (Nafziger, 1994; Ahmadi et al., 1993; Hashemi-Defzfouli and Herbert, 1992; Thomison and Jordan, 1995). Tollenaar (1989) reported higher optimum plant populations for hybrids released in the 1980s than for hybrids released in the 1960s and attributed the difference to improved drought tolerance.

The availability of residual soil-applied herbicides and increased cold tolerance in corn hybrids have enabled producers to use earlier planting dates. Corn producers in northeast Kansas can plant earlier to increase growing season length or move harvest earlier in the year to distribute labor. Approximately 260 heat units can be accumulated during April in northeast Kansas, yet few data exist to quantify the effects that these additional heat units have on corn yields. Swanson and Wilhelm (1996) reported an optimum planting date of 12 May for Lincoln, NE, whereas the optimum planting date was 24 April for Dekalb, IL (Nafziger, 1994). Ahmadi et al. (1993) reported yield losses as a result of delayed planting in Ohio.

Approximately 440 000 acres of corn were planted in northeast and north central Kansas in 1995, with the north central area reporting a 25% increase in corn acres since 1990 (Kansas Farm Facts, 1990, 1996). As corn acres increase in Kansas, management decisions must be modified to fit the current hybrids and environmental conditions. The objectives of this study were to evaluate the effects of plant population, planting date, and hybrid maturity on corn yield.

MATERIALS AND METHODS

Six field studies were conducted at three dryland locations in Kansas in 1994 and 1996 to assess the impacts of

Table 1. Planting dates and monthly rainfall and heat unit totals for six location-years in Kansas.

Location years	Planting dates	Rainfall April	May	June	July	August	September	Total	Heat units† April	May	June	July	August	September	Total
					in.							50°F			
Powhattan															
1994	4 April, 5 May, 5 June	4.4	3.1	3.8	4.0	1.3	1.4	18.1	251	499	710	734	687	529	3410
1996	4 April, 3 May, 10 June	1.8	7.7	3.9	5.4	5.5	3.1	27.4	236	401	713	778	709	455	3291
Normal		3.1	4.2	5.4	4.1	4.2	4.5	25.5	268	470	702	809	768	508	3528
Rossville															
1994	4 April, 7 May, 8 June	4.1	1.3	6.8	2.6	7.6	1.3	23.7	287	479	743	759	754	506	3528
1996	5 April, 1 May, 13 June	0.7	3.1	4.2	2.3	5.1	2.3	17.8	289	512	735	803	722	478	3538
Normal		3.2	3.9	5.3	4.0	3.6	3.4	23.4	259	450	737	855	769	550	3620
Scandia															
1994	5 April, 5 May, 9 June	2.0	1.5	8.5	5.6	0.6	1.0	19.2	243	513	708	723	717	541	3445
1996	8 April, 3 May, 13 June	2.8	5.1	1.0	5.4	3.7	4.1	22.1	251	397	704	772	693	458	3275
Normal		2.4	3.7	4.8	3.3	3.3	3.5	20.9	240	448	703	823	786	502	3502

† Heat units calculated with $T_{max} \leq 86$ and $T_{min} \geq 50$.

planting date, plant population, and hybrid maturity on corn yield. These studies were conducted on a Crete silt loam soil (fine, mixed, mesic Typic Argiudoll) at the Irrigation Experiment Field near Scandia; a Grundy silt loam soil (fine, mont morillonic, mesic Aquertic Arqiudoll) at the Cornbelt Experiment Field near Powhattan; and a Eudora silt loam (course-silty, mesic, Fluventic Hapludolls) at the Kansas River Valley Experiment Field near Rossville.

Two corn hybrids, Golden Harvest H-2404 (102-d relative maturity) and Golden Harvest H-2573 (113-d relative maturity) were planted each year in early April, May, and June (Table 1). Target populations were 14 000, 20 000, and 26 000 plants/acre. Plots were not thinned, so seeding rates that were 115% of the target population were used in an attempt to achieve the desired final stands. On average, the final plant stands were 15 966, 21 100 and 25 395 plants/acre for the respective treatments.

Corn was the previous crop in all experiments. Soil fertility levels were amended to ensure that nutrient availability would not be a limiting factor. Weed control was accomplished by a preemergence application of atrazine at a rate of 1.6 lb/acre and metolachlor at a rate of 2 lb/acre followed by postemergence applications of either primisulfuron (0.018 lb/acre) and prosulfuron (0.018 lb/acre) or bromoxynil (0.25 lb/acre) and atrazine (0.50 lb/acre).

A split-plot arrangement of treatments in a randomized complete block experimental design with three replicates was used. Planting date was the main plot, and plant population and hybrid maturity were the subplots. Sub-plots consisted of four 30-ft rows in 30-in spacing. The center two rows of these four-row plots were machine harvested at physiological maturity, and all grain was adjusted to 15.5% moisture to determine yields. Final plant stands were determined at harvest.

Variances for grain yields for each location-year were tested for homogeneity using Hartley's test for equal variances (Ott, 1988). Homogeneous variances of grain yields did not exist for the six location-years, nor for individual years at each location. Therefore, yield data were not combined and analysis of variance was used to determine treatment effects on grain yield at each individual location-year. Mean separations were accomplished using an LSD (α = 0.05).

RESULTS

The six location-years of this study experienced a wide range of climatic conditions, which are typical in Kansas (Table 1). As a result of this variation, planting date consistently influenced grain yields across all six location-years as either a main effect or part of an interaction ($P < 0.05$) (Table 2). Plant population and hybrid maturity influenced yields at five of the six location-years.

Powhattan

In 1994 at Powhattan, both planting date and plant population significantly influenced grain yield (Table 2). Grain yields declined 8 bu/acre as planting was delayed from 4 April until 5 May (Table 3) and an additional 11 bu/acre when planting was delayed until 5 June. Grain yields increased 8 bu/acre as plant population increased from 14 000 to 20 000 plants/acre (Fig 1). Increasing plant population from 20 000 to 26 000 plants/acre did not further increase yields.

At Powhattan in 1996, the three-way interaction among planting date, hybrid maturity, and plant population was significant ($P < 0.01$), because H-2404 responded more erratically to the increasing plant population, and H-2573 produced lower yields when planted in June. (Fig. 2). Because of the difficulty in discussing a three-way interaction, it will be discussed as the two-way interaction of most interest (planting date × plant population) at each level of the main effect of least interest (hybrid maturity) as outlined by Milliken and Johnson (1984).

When H-2404 was used, there was no grain yield response to the plant population treatments with the 4 April planting date (Fig. 2A). With the 3 May planting date, yield of the 14 000 plants/acre treatment was more than 43 bu/acre lower than yield of the other two plant populations. With the 10 June planting date, grain yields from the 14 000 and 26 000 plants/acre treatments were approximately 17 bu/acre lower than those from the 20 000 plants/acre treatment.

When the later maturing H-2573 was used, grain yields from the 20 000 plants/acre treatment were greater than those from the 14 000 plants/acre treatment and equal to yields from the 26 000 plants/acre treatment with the 4 April

Table 2. Statistical F-test probabilities for corn grain yields at six location-years.

Source	Powhattan 1994	Powhattan 1996	Rossville 1994	Rossville 1996	Scandia 1994	Scandia 1996
	Prob > F					
Date	0.01	0.01	0.07	0.01	0.01	0.01
Hybrid	0.09	0.08	0.01	0.01	0.01	0.01
Population	0.01	0.01	0.01	0.01	0.07	0.01
Hyb × Pop	0.49	0.07	0.02	0.89	0.63	0.23
Date × Hyb	0.57	0.29	0.01	0.01	0.06	0.03
Date × Pop	0.77	0.01	0.01	0.09	0.08	0.17
D × H × P	0.70	0.01	0.90	0.04	0.74	0.18
C.V. (%)	8.24	9.09	11.76	13.78	10.61	12.70

planting date (Fig 1B). Grain yields did not differ between the 14 000 and 26 000 plants/acre treatments.

With the 3 May planting date, increasing plant population from 14 000 to 20 000 plants/acre and from 20 000 to 26 000 plants/acre resulted in no differences in grain yields. Yields increased 19 bu/acre, however as the plant population increased from 14 000 to 26 000 plants/acre. There was no yield response to increasing plant population with the 10 June planting date.

Rossville

In 1994 at Rossville, the influence of hybrid maturity and plant population varied as planting was delayed as indicated by the significant interactions for planting date by hybrid maturity ($P < 0.01$) and planting date × plant population ($P < 0.01$) (Table 2). Growing season length influenced hybrid yields in 1994. Growing season length for plantings in April or May was adequate for the later maturing H-2573 to exhibit its higher yield potential (Table 4). The yield differences between the two hybrids were 64 bu/acre with the 4 April planting date and 22 bu/acre with the 7 May planting date. However, when planting was

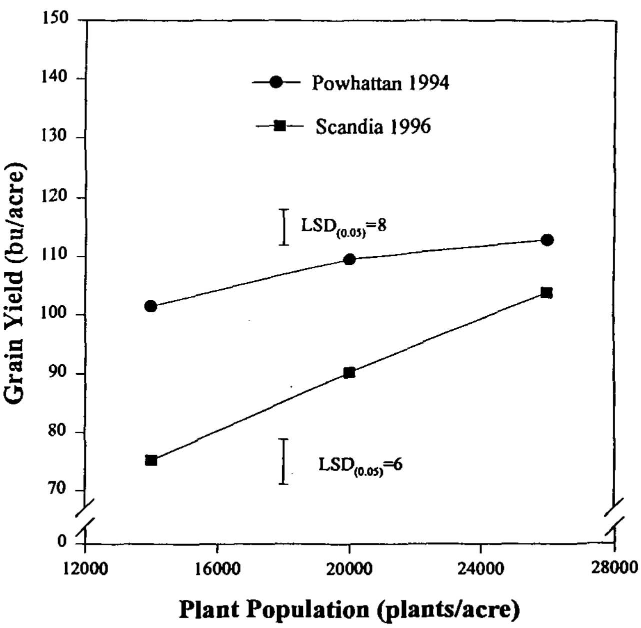

Fig. 1. Corn yields with three plant populations at Powhattan in 1994 and Scandia in 1996.

Table 3. Corn grain yields as affected by planting date at Powhattan, KS, in 1994.

Planting date	Grain yield
	Powhattan 1994
	bu/acre
4 April	117
4 May	109
5 June	90
$LSD_{(0.05)}$	6

delayed until 8 June, H-2404 produced 29 bu/acre more grain than H-2573.

Increasing plant population increased grain yields with all three planting dates in 1994 (Table 4). With the 4 April planting date, increasing plant population from 14 000 to 20 000 plants/acre did not increase grain yields, but increasing plant population from 20 000 to 26 000 plants/acre resulted in a 15 bu/acre yield increase. Increasing plant population from 14 000 to 20 000 plants/acre resulted in a 52 bu/acre yield increase with the 7 May planting date. A 17 bu/acre yield increase occurred when plant population was increased from 20 000 to 26 000 plants/acre with this planting date. With the 8 June planting date, grain yields from 14 000 and 20 000 plants/acre were not different. There were no difference in grain yields from 20 000 and 26 000

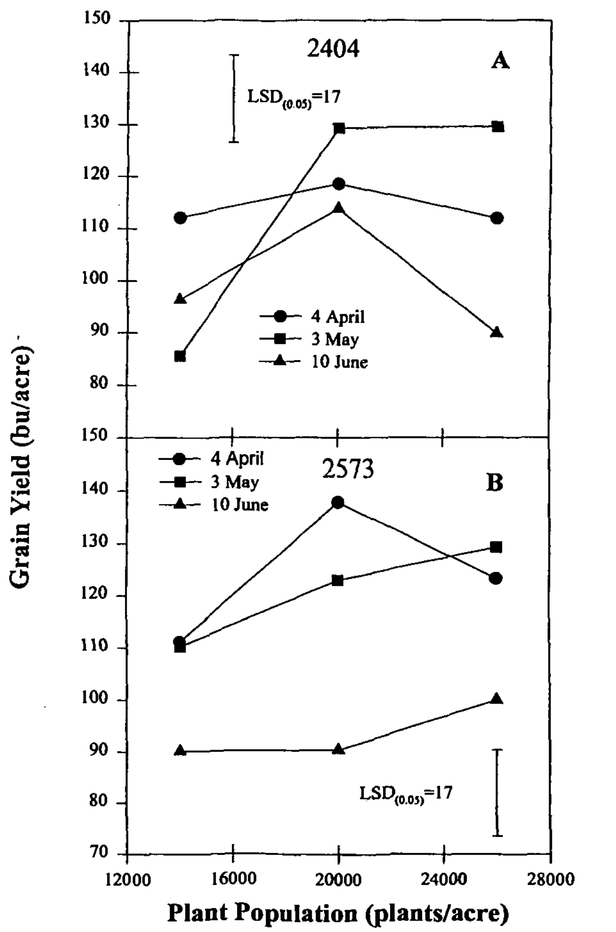

Fig. 2. Grain yields for two corn hybrids with three planting dates and three plant populations at Powhattan in 1996.

Table 4. Corn grain yields as affected by planting date, plant population, and hybrid at Rossville, KS, 1994.

Plant population	Planting date		
	4 April	7 May	8 June
	bu/acre		
Plants/acre			
14 000	115	75	67
20 000	127	127	79
26 000	142	144	89
$LSD_{(0.05)}$		15†	
Hybrid			
H-2404	93	100	93
H-2573	157	122	64
$LSD_{(0.05)}$		12†	

† LSD for grain yield comparisons within a given planting date only.

plants/acre; however, grain yields from 14 000 plants/acre were 22 bu/acre lower than yields from 26 000 plants/acre.

The three-way interaction among planting date, hybrid maturity, and plant population was significant ($P < 0.05$) in 1996 (Table 2). This three-way interaction existed because grain yields were reduced when planting was delayed until June in H-2573, and no significant yield response to plant population occurred with the June planting date for either hybrid (Fig. 3).

For both hybrids, delaying planting from 5 April to 1 May had little effect on yield (Fig. 3). This response was the

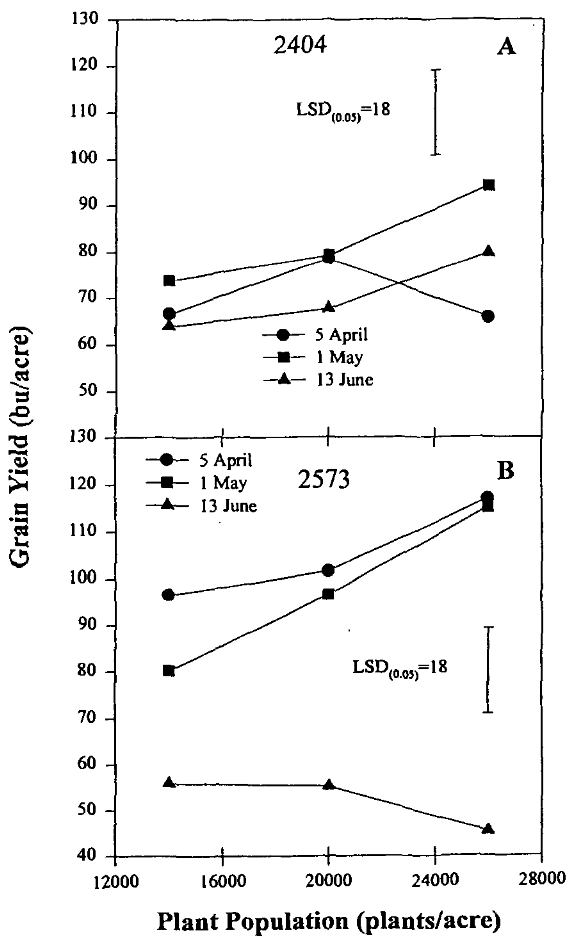

Fig. 3. Grain yields for two corn hybrids with three planting dates and three plant populations at Rossville in 1996.

result of below-normal rainfall in April, which reduced any growth advantage that might have been gained by planting a month early. However, delaying planting from 1 May until 13 June reduced yields 44 bu/acre for H-2573, yet reduced grain yields by only 12 bu/acre in H-2404. The 1996 growing season was above average in length, with below-normal rainfall from April through July. This resulted in the later maturing H-2573 producing yields that were approximately 11 bu/acre greater than those of H-2404. Under these conditions, a greater portion of the early reproductive period for H-2404 occurred under stress, thus reducing its yields.

When H-2404 was used, increasing plant population from 14 000 to 26 000 plants/acre did not affect grain yields with the 5 April planting date (Fig 3A). As mentioned, rainfall was below normal at this location in 1996, thus limiting early-season growth and yield potential. Karlen and Camp (1985) indicated that at lower yield potentials, grain yields did not respond to increasing plant population. This lack of response can be attributed to the fact that adequate plant numbers existed at the lower population to produce enough kernels on an area basis to produce the yield that the environment can support.

The crop planted on 1 May was able to take advantage of rainfall received in late May and early June, resulting in a yield response to increasing plant population. Increasing plant population from 14 000 to 20 000 plants/acre did not increase grain yields. However, increasing plant population from 20 000 to 26 000 plants/acre increased grain yields 21 bu/acre (Fig. 3A).

In August and September, grain fill occurred under cooler temperatures for the crop planted on 13 June than the corn planted on the other two dates, and thus, lower grain yields. As with the 5 April planting date, the lower overall yield level resulted in no yield response from increasing plant population.

When H-2573 was used, the higher yield potential of this later maturing hybrid resulted in increased yields from increasing plant population with the 5 April planting date (Fig 3B). Grain yields did not increase when plant population increased from 14 000 plants/acre to 20 000 plants/acres. However, grain yields from 14 000 plants/acre were approximately 20 bu/acre lower than yields from 26 000 plants/acre. Grain yields for H-2573 followed a similar pattern at the later two planting dates in 1996. Grain yield increased as plant population increased from 14 000 to 26 000 plants/acre with the 1 May planting, but no plant population response occurred with the 13 June planting.

Scandia

At Scandia in 1994, hybrid maturity and planting date affected grain yields ($P < 0.01$) (Table 2). Unlike results at the other two locations, the short-season H-2404 produced approximately 22 bu/acre more grain than H-2573 (Table 5). This difference in the two hybrids was the result of a different rainfall pattern. Above-average rainfall in June and July and below-average rainfall in August and September favored the short-season H-2404 compared with H-2573. Below-average heat unit accumulation in August and September also may have delayed development in H-2573, so that grain fill occurred under cooler temperatures.

Table 5. Corn grain yields as affected by hybrid and planting date at Scandia, KS, 1994.

Hybrid	Grain yield
	bu/acre
H-2404	174
H-2573	152
$LSD_{(0.05)}$	10
Planting date	
5 April	200
5 May	193
9 June	95
$LSD_{(0.05)}$	12

The impacts of growing season temperatures also were illustrated in the effect that planting date had upon yields in 1994 (Table 5). Grain yields with the 5 April and 5 May planting dates were above 190 bu/acre, whereas delaying planting until 9 June reduced grain yields by approximately 100 bu/acre. Heat unit accumulation was approximately 200 heat units below normal from June through September, thus forcing late development and grain fill under cooler temperatures with the 9 June planting date than the earlier planting dates.

The effects of hybrid maturity on grain yield varied across the three planting dates as indicated by the significant two-way interaction ($P < 0.05$) (Table 2). Plant population also influenced grain yields in 1996. Grain yields increased 15 bu/acre when plant population was increased from 14 000 to 20 000 plants/acre (Fig. 1). When the plant population increased another 6 000 plants/acre, grain yields increased an additional 14 bu/acre.

The differences between the two hybrids and three planting dates were primarily the result of the hybrid response at the first two planting dates (Table 6). With an 8 April planting date, grain yields for H-2404 were 26 bu/acre lower than those for H-2573. The difference between the two hybrids was 39 bu/acre with the 3 May planting date. Below-average rainfall in June resulted in overall lower yields with the first planting date, as well as the hybrid difference. This drought occurred at the time of tassel emergence in the April-planted crop and appeared to have a bigger impact on H-2404, indicating that it is either weaker in drought tolerance than H-2573 or that the stress symptoms were relieved before H-2573 reached that stage. With the 13 June planting date, grain yields from the two hybrids were not different.

DISCUSSION

Overall, planting date had a greater impact on grain yields than plant population. At Powhattan and Rossville, planting corn in early April resulted in yields that were 5 bu/acre greater than those for corn planted in early May, when averaged across hybrids and plant population treatments. However, when planting was delayed until early June, grain yields were 23 bu/acre lower than those with the May planting date.

These early planting dates enabled the crop to make use of the additional heat units that were accumulated in April, thus extending the growing season. These results agree with Ahmadi et al. (1993), who reported that at several sites in Missouri, maximum yields were produced with the earliest planting date, and yields declined as planting was delayed.

Table 6. Corn grain yields as affected by hybrids and planting date at Scandia, KS, 1996.

Planting date	Hybrid H-2404	Hybrid H-2573
	bu/acre	
8 April	46	72
3 May	99	138
13 June	90	94
$LSD_{(0.05)}$	11†	

† LSD for grain yield comparisons within a given hybrid only.

At Scandia, delaying planting from April to May had different effects each year. In 1994, no yield differences occurred between the first two planting dates. In 1996, however, grain yields increased by 60 bu/acre when planting was delayed from early April until early May. This response was the result of drought and heat stress in late June and early July. The April-planted crop encountered this stress during early reproductive stages. Gordon et al. (1995) reported maximum corn yields when irrigation treatments were applied at tassel emergence and 1 and 2 wk later. Classen and Shaw (1970), Hall et al. (1981), and Nesmith and Ritchie (1992) associated yield losses from water stress with a reduction in the number of well developed kernels. Grant et al. (1989) indicated that final kernel number was reduced by water stress that began 7 d prior to silking and continued until 21 d after silking. They also found that the greatest reduction in kernel number occurred when water stress occurred 7 d after silking. The events that occurred in 1996 illustrate some of the risks associated with dryland corn production in north central Kansas.

Delaying planting from early May until early June at Scandia reduced yields by 56 bu/acre. This yield reduction was the result of grain fill occurring later in the growing season. Average heat unit accumulation across all three locations was 529 in September compared with 755 in August. Cooler temperatures reduce grain fill rates on a daily basis as well as extend the length of grain fill. Extending this process increases the risk of a freeze occurring prior to physiological maturity.

These results support the findings of Swanson and Wilhelm (1996), who reported that delaying planting from early May until early June at Lincoln, NE, reduced grain yields by approximately 49 bu/acre. They reported an optimal planting date of approximately 12 May. Based on these results, the optimal planting date for Scandia, KS, would be approximately 1 May.

Grain yield response to hybrid maturity varied consistently with planting dates across all three locations (Table 1). As expected, H-2573 produced higher yields than the earlier maturing H-2404 when planted in April. On average across all six location-years, this yield difference was 23 bu/acre. When planted in May, the difference between the two hybrids was reduced to approximately 4 bu/acre. When planted in June, the short-season hybrid produced yields that averaged 5 bu/acre higher than those from the full-season H-2573.

A planting date × hybrid maturity interaction was expected, because delaying planting until June often results in delays in maturity, thus forcing grain development to occur under cooler temperatures in August and September.

These results indicate that if early planting dates are used, the producers should take advantage of the longest-season hybrid adapted for the area. If planting is delayed to early June, using a shorter-season corn hybrid such as H-2404 appears to have an advantage. In half of the six location-years, H-2404 produced less grain than H-2573 when planted in June. However, the average yield difference was only 4 bu/acre, compared with an average yield difference of 22 bu/acre in the location-years when H-2404 produced yields greater than H-2573 from a June planting date.

Across all six location-years, increasing plant population increased grain yields. A final population of 26 000 plants/acre resulted in an average yield of 117 bu/acre, whereas a population of 20 000 plants/acre resulted in an average yield of 112 bu/acre. Over these same environments, 14 000 plants/acre resulted in an average yield of 99 bu/acre.

Higher plant populations increased grain yield by increasing the potential and final kernel number per acre. Kernel numbers per acre were determined at Powhattan and Rossville in 1996 and accounted for 85% of the variation in grain yield (data not shown). Analysis of previous research (Hashemi-Dezfouli and Herbert, 1992; Whigham and Woolley, 1974; Karlen and Camp, 1985; Shumway et al., 1992; Jama and Ottman, 1993; NeSmith and Ritchie, 1992; Swanson and Wilhelm, 1996; Ahmadi et al., 1993) also indicated a strong relationship between final grain yield and kernels per acre ($r^2 = 0.89$).

These results indicate that because kernel number per acre accounted for a larger amount of variation in grain yield than kernel weight, increasing plant population should result in the maximum kernel number per acre. At higher plant population under excellent growing conditions, the additional ears per acre will result in maximum grain yield by maximizing kernels per acre. At the lower plant population under these same growing conditions, the genetic limit on ear size and subsequent kernels per acre may reduce yields, because the crop cannot produce enough kernels to meet the environmental potential.

Karlen and Camp (1985) reported that under moisture-limiting conditions, no yield differences occurred between low and high plant populations, because both populations were adequate to produce enough kernels to produce the yield that the environment would support. However, when irrigation was applied to relieve moisture stress, higher plant population resulted in higher yields than the low plant population. As the yield potential of the environment increased, higher plant population was required to produce enough kernels per acre to meet that potential.

These results also indicate that as drought and heat tolerance improve in commercial hybrids, the optimal plant population and yield level will increase. Tollennar (1989) reported that the optimum plant population for hybrids released in the 1980s was higher than for hybrids released around 1960. Nissanka et al. (1997) reported that this increase was the result of better moisture stress tolerance and more efficient carbon assimilation.

The results from Scandia in 1996 support their findings. Dry weather and high temperatures prevailed in late June and early July, when corn planted in April was beginning to tassel. Despite the fact that the environmental stress occurred during a critical growth stage, the higher plant population did not result in reduced yields compared with the lowest plant population.

CONCLUSIONS

Planting corn in early April and early May resulted in similar grain yields, at five of the six location-years. Higher temperatures and lower rainfall amounts in June reduced grain yields with an April planting compared with a May planting date at one location-year. Increasing corn plant population increased grain yields under a majority of the environments, but reduced grain yields under one set of conditions. In general, yields of the short-season hybrid were greater than or equal to those of the full-season hybrid at the later planting dates. The full-season hybrid generally produced more grain than the short-season hybrid when planted early and growing season length was not a yield-limiting factor.

REFERENCES

Ahamadi, M., W.J. Wiebold, J.E. Beuerlein, D.J. Eckert, and J. Schoper. 1993. Agronomic practices that affect corn kernel characteristics. Agron. J. 85:615–619.

Classen, M.M., and R.H. Shaw. 1970. Water deficit effects on corn. II. Grain components. Agron. J. 62:652–655.

Gordon, W.B., R.J. Raney, and L.R. Stone. 1995. Irrigation management practices for corn production in north central Kansas. J. Soil Water Conserv. 50:395–398.

Grant, R.F., B.S. Jackson, J.R. Kiniry, and G.F. Arkin. 1989. Water deficit timing effects on yield components in maize. Agron. J. 81:61–65.

Hall, A.J., J.H. Lemcoff, and N. Tapani. 1981. Water stress before and during flowering in maize and its effects on yield, its components and the determinants. Maydica 26:19–38.

Hashemi-Dezfouli, A., and S.J. Herbert. 1992. Intensifying plant density response of corn with artificial shade. Agron. J. 84:547–551.

Jama, A.O., and M.J. Ottman. 1993. Timing of the first irrigation in corn and water stress conditioning. Agron. J. 85:1159–1164.

Kansas Farm Facts 1990. 1991. Kansas Dep. of Agriculture, Topeka.

Kansas Farm Facts 1996. 1997. Kansas Dep. of Agriculture, Topeka.

Karlen, D.L., and C.R. Camp. 1985. Row spacing, plant population, and water management effects on corn in the Atlantic coastal plain. Agron. J. 77:393–398

Milliken, G.E., and D.E. Johnson. 1984. Analysis of messy data. Vol. 1. Lifetime Learning Publ. Belmont, CA.

Nafziger, E.D. 1994. Corn planting date and plant population. J. Prod. Agric. 7:59–62.

NeSmith, D.S., and J.T. Ritchie. 1992. Short- and long-term responses of corn to a pre-anthesis soil water deficit. Agron. J. 84:107–113.

Nissanka, S.P., M.A. Dixon, M.Tollenaar. 1997. Canopy gas exchange response to moisture stress in old and new maize hybrids. Crop Sci. 37:172–181.

Ott, L. 1988. An introduction to statistical methods and data analysis. 3rd ed. PWS-Kent Publishing, Boston.

Shumway, C.R., J.T. Cothren, S.O. Serna-Saldivar, and L.W. Rooney. 1992. Planting date and moisture effects on yield, quality, and alkaline-processing characteristics of food-grade maize. Crop Sci. 32:1265–1269.

Swanson, S.P., and W.W. Wilhelm. 1996. Planting date and residue rate effects on growth, partitioning, and yield of corn. Agron. J. 88:205–210.

Thomison, P.R., and D.M. Jordan. 1995. Plant population effects on corn hybrids differing in ear growth habit and prolificacy. J. Prod. Agric. 8:394–400.

Tollenaar, M. 1989. Genetic improvement in grain yield of commercial maize hybrids grown in Ontario from 1959 to 1988. Crop Sci. 29:1365–1371

Vanderlip, R.L. 1968. How plant populations affect yields of corn hybrids. Kansas Agric. Exp. Stn. Bull. 519.

Whigham, D.K., and D.G. Woolley. 1974. Effect of leaf orientation, leaf area, and plant densities on corn production. Agron. J. 66:482–486.